

Project 4: Multi-Agent Brand Health Monitoring & Insight Synthesis

C5i-style project sheet converted into a readable PDF format

Field	Details
Domain / Use-Case Areas	Marketing Analytics; Brand Measurement; Campaign Analytics; Social Media Intelligence; Search Analytics; Consumer & Market Insights
Technology Stack / Themes	CrewAI, app.crew, Azure AI Foundry agents or equivalent low-code/no-code agents, agentic workflow orchestration, tool calling, critique agent, planner-executor pattern, HITL
Project Description	Design a multi-agent brand health monitoring system for a CPG, retail, or digital brand team. Specialized agents should collect signals from social listening, search trends, campaign performance, reviews, and competitor activity; synthesize weekly brand health insights; detect anomalies; and produce an executive-ready insight brief.
Input Data / Assets	Synthetic social posts, search trend time series, campaign metrics, review exports, competitor news snippets, brand tracker summaries, weekly KPI dashboards
Evaluation Metrics	Tool-call accuracy, trace completeness, insight relevance, anomaly detection precision, contradiction detection, executive brief quality, cost/latency per run, analyst time saved

High-Level Work Areas

- Define brand-health KPIs: awareness, sentiment, share of voice, campaign lift, search interest, review themes, and competitor movement.
- Create specialist agents: Social Listening Agent, Search Trend Agent, Campaign Agent, Review Theme Agent, Competitor Agent, Insight Synthesizer, and Critic/QA Agent.
- Connect tools or mock APIs for each signal source.
- Build an orchestration flow where agents produce evidence cards and the synthesizer creates the final brief.
- Add critique, contradiction checks, source quality scoring, and human review.
- Compare single-agent summarization with multi-agent decomposition.
- Track tool usage, cost, latency, and trace completeness.

Progressive Demo Sequence

- Start with the executive question: 'What changed in brand health this week and what should marketing do?'
- Show a single LLM summarizing all inputs in one prompt. Highlight context overload, weak traceability, and missed channel-specific signals.
- Add a simple sequential workflow: ingest social, reviews, search, and campaign data. Show improved organization but no specialized reasoning.
- Add specialist agents: Social Listening, Search Trend, Campaign, Review Theme, and Competitor Agent. Each produces an evidence card.
- Ask each agent targeted questions, e.g., 'What negative themes spiked?', 'Which keywords rose?', 'Did the campaign move consideration?', 'What competitor action changed?'
- Add an Insight Synthesizer agent that combines evidence into a weekly brand-health narrative.
- Add a Critic/QA agent that checks whether claims have evidence, whether channels contradict each other, and whether recommendations follow from the data.
- Add HITL: brand manager approves/edits the final executive brief and marks weak insights.
- Show observability: agent traces, tool calls, failed calls, latency, and cost.
- Final limitations to highlight: agents can amplify poor input data, tool/API failures can break workflow, agent loops increase cost, and recommendations may still need causal validation.
- Future scope: automated anomaly detection, MMM/MTA integration, campaign simulation, brand risk alerts, memory across weeks, and production-grade AgentOps monitoring.